

Student Number: _____ Class Teacher: _____

St George Girls High School

Trial Higher School Certificate Examination

2017



Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 15, show relevant mathematical reasoning and/or calculations

Section I	/10
Section II	
Question 11	/12
Question 12	/12
Question 13	/12
Question 14	/12
Question 15	/12
Total	/70

Total Marks – 70

Section 1

Pages 3 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Answer on the multiple choice answer sheet provided at the back of this paper

Section II

Pages 7 – 11

60 marks

- Attempt Questions 11 – 15
- Allow about 1 hour and 45 minutes for this section
- Begin each question in a new writing booklet

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Section I

10 marks

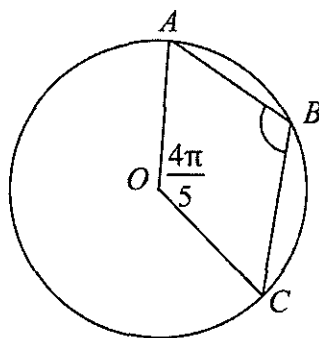
Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

- 1 The points A , B and C lie on a circle with centre O , as shown in the diagram.

The size of $\angle AOC$ is $\frac{4\pi}{5}$ radians.



Not to scale

What is the size of $\angle ABC$ in radians?

- (A) $\frac{3\pi}{10}$
- (B) $\frac{\pi}{2}$
- (C) $\frac{3\pi}{5}$
- (D) $\frac{4\pi}{5}$
- 2 Which of the following is the exact value of $\int_{\frac{3}{\sqrt{2}}}^3 \frac{4 \, dx}{\sqrt{9-x^2}}$?
- (A) $-\pi$
- (B) $-\frac{\pi}{4}$
- (C) $\frac{\pi}{4}$
- (D) π

- 3 What are the coordinates of the point that divides the interval joining $P(2, 1)$ and $Q(2, 8)$ internally in the ratio 3: 4?
- (A) $(1, 7)$
(B) $(2, 4)$
(C) $(2, 7)$
(D) $(4, 2)$
- 4 An oil slick is in the shape of a circle. Its surface area is increasing at a rate of $10 \text{ m}^2/\text{s}$. Let r metres be the radius of the oil slick in t seconds.

The rate of increase of r in m/s , is given by

- (A) $\frac{5}{\pi r}$
(B) $\frac{20}{\pi r}$
(C) $\frac{10}{\pi r^2}$
(D) $\frac{1}{20\pi r}$
- 5 Let $f(x) = \frac{2}{x-3} + 1$.

The equations of the asymptotes of the graph of the inverse function $f^{-1}(x)$ are

- (A) $x = 1$ and $y = 3$
(B) $x = 1$ and $y = -3$
(C) $x = 3$ and $y = 1$
(D) $x = -3$ and $y = -1$

- 6 A particle moves in a straight line such that it's displacement from the origin is x metres.

The velocity of the particle at any point is given by $v = 2x^2 - 3$ m/s.

Find the acceleration of the particle when it is 2 units to the right of the origin.

(A) $-\frac{2}{3}$ m/s²

(B) 5 m/s²

(C) 8 m/s²

(D) 40 m/s²

- 7 The function $f(x) = \sin x - \frac{2x}{3}$ has a real root close to $x = 1.5$.

Let $x = 1.5$ be a first approximation to the root.

What is the second approximation to the root using Newton's method?

(A) 1.495

(B) 1.496

(C) 1.503

(D) 1.504

- 8 If $\sqrt{3} \tan x = -1$ which expression gives all the possible values of x , where n is an integer?

(A) $x = 2n\pi \pm \frac{\pi}{6}$

(B) $x = n\pi - \frac{5\pi}{6}$

(C) $x = n\pi + \frac{\pi}{6}$

(D) $x = n\pi - \frac{\pi}{6}$

9 What is the domain and range of $y = 3 \sin^{-1}(2x)$?

- (A) Domain: $-\frac{1}{2} \leq x \leq \frac{1}{2}$. Range $-\frac{1}{3} \leq y \leq \frac{1}{3}$
(B) Domain : $-2 \leq x \leq 2$. Range $-\frac{1}{3} \leq y \leq \frac{1}{3}$
(C) Domain : $-\frac{1}{2} \leq x \leq \frac{1}{2}$. Range $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$
(D) Domain : $-2 \leq x \leq 2$. Range $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

10 The roots of $2x^3 - 6x^2 - 8x + 12 = 0$ are α , β and γ .

What is the value of $(\alpha + 2)(\beta + 2)(\gamma + 2)$?

- (A) -12
(B) -6
(C) 6
(D) 12

End of Section I

Section II

60 marks

Attempt Questions 11 – 15

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (12 marks) Use a separate writing booklet	Marks
(a) Find the size of the acute angle between the lines $x - y - 4 = 0$ and $3x - y + 4 = 0$. Answer to the nearest degree.	2
(b) Differentiate $y = \log_e(\sin^{-1}x)$	2
(c) Find $\int \frac{1}{9+x^2} dx$	2
(d) Evaluate $\lim_{x \rightarrow 0} \frac{3x}{\sin 2x}$	2
(e) Show that $(x + 3)$ is a factor of $x^3 - 3x^2 - 10x + 24$ and hence factorise $x^3 - 3x^2 - 10x + 24$ fully.	4

End of question 11

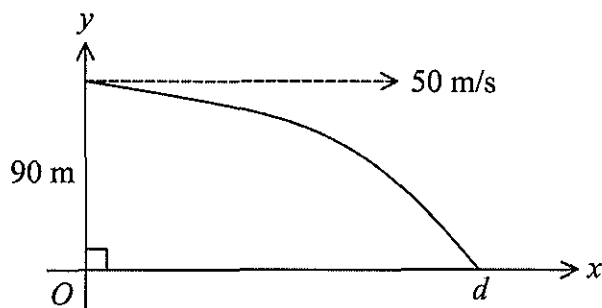
Question 12 (12 marks) Use a separate writing booklet **Marks**

(a) Evaluate $\int_{-1}^0 x\sqrt{1+x} \, dx$, using the substitution $u = 1 + x$ **3**

(b) Solve the inequality $\frac{2x}{x-1} \geq 1$ **3**

(c) Find the exact value of $\sin(2 \tan^{-1} \frac{1}{2})$ **2**

- (d) The diagram below shows the trajectory of a ball thrown horizontally, at a speed of 50 ms^{-1} , from the top of a tower 90 metres above ground level.



The ball strikes the ground d metres from the base of the tower.

- (i) Show that the equations describing the trajectory of the ball are: **2**

$$x = 50t \text{ and } y = 90 - \frac{1}{2}gt^2$$

where g is the acceleration due to gravity and t is the time in seconds.

- (ii) Prove that the ball strikes the ground at time $t = 6\sqrt{\frac{5}{g}}$ seconds. **1**

- (iii) How far from the base of the tower does the ball strike the ground? **1**

End of question 12

Question 13 (12 marks) Use a separate writing booklet

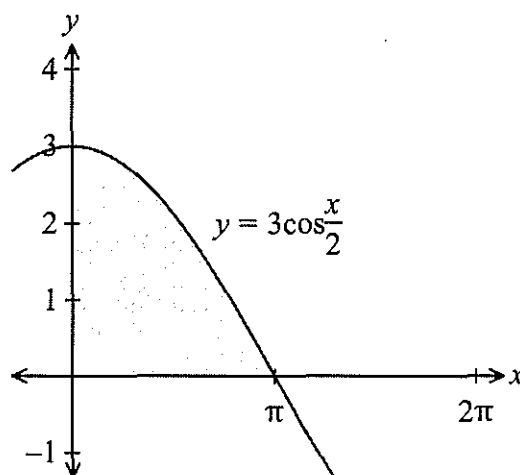
Marks

- (a) (i) Express $\sqrt{3}\sin t + \cos t$ in the form $R\sin(t + \alpha)$ where α is in radians, 2
and $0 \leq \alpha \leq \frac{\pi}{2}$

- (ii) Hence, or otherwise, find the solutions of the equation 2

$$\sqrt{3}\sin t + \cos t = \sqrt{3} \quad \text{for } 0 \leq t \leq 2\pi$$

- (b) The region bounded by the graph and the x -axis between and is rotated about the x -axis to form a solid 3



Find the exact volume of the solid.

- (c) Newton's law of cooling states that when an object at temperature $T^\circ\text{C}$ is placed in an environment at temperature $T_0^\circ\text{C}$, the rate of the temperature loss is given by the equation $\frac{dT}{dt} = -k(T - T_0)$ where t is the time in minutes and k is a positive constant.

- (i) Show that $T = T_0 + Ae^{-kt}$ satisfies the above equation. 2

- (ii) An object whose initial temperature is 60°C is placed in a room in which the internal temperature is maintained at 12°C . After 25 minutes, the temperature of the object is 30°C . 3
How long will it take for the object's temperature to reduce to 15°C ?

End of question 13

Question 14 (12 marks) Use a separate writing booklet **Marks**

- (a) $P(2at, at^2)$ is any point on the parabola $x^2 = 4ay$. The line d is parallel to the tangent at P and passes through the focus S of the parabola.
- (i) Show the equation of the tangent at P is $y = tx - at^2$ 1
- (ii) Find the equation of the line d . 1
- (iii) The line d intersects the x -axis at the point R .
 Find the coordinates of the midpoint, M , of the interval RS . 2
- (iv) Find the equation of the locus of M . 1
- (b) Simplify $\cos(2\cos^{-1}x)$ and hence evaluate $\int_0^{\frac{1}{2}} \cos(2\cos^{-1}x) dx$. 2
- (c) A particle is moving in a straight line performing Simple Harmonic Motion.
 At time t seconds it has a displacement x metres from a fixed point O on the line given by
- $$x = 1 + 2\cos\left(2t - \frac{\pi}{3}\right)$$
- (i) Show that $\ddot{x} = -4(x - 1)$ 1
- (ii) Find the centre of the motion and the time taken for the particle to first reach maximum speed. 2
- (iii) Find the first time the particle is at rest and the amplitude of the motion 2

End of question 14

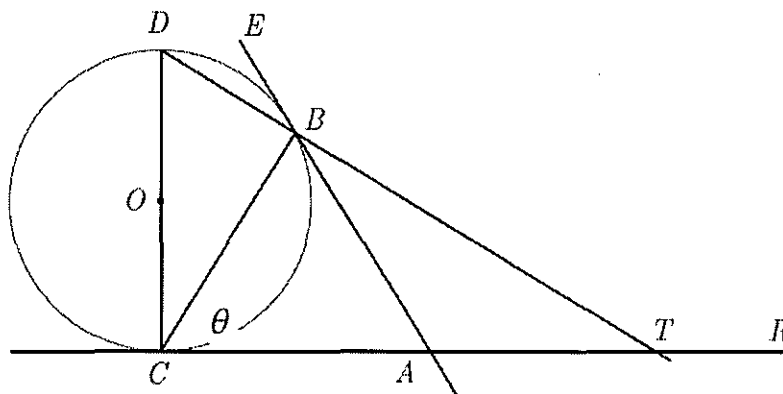
Question 15 (12 marks) Use a separate writing booklet

Marks

- (a) Use Mathematical Induction to show that $5^n > 4^n + 3^n$
 for all integers $n \geq 3$

3

(b)



In the diagram, CD is the diameter of the circle, centre O, and CR is a tangent to the circle C.

The line DT intersects the circle at B and CR at T.

The tangent to the circle at B intersects CR at A and $\angle BCA = \theta$.

Copy this diagram into your examination booklet.

- (i) Prove that $\angle ABT = \frac{\pi}{2} - \theta$.

2

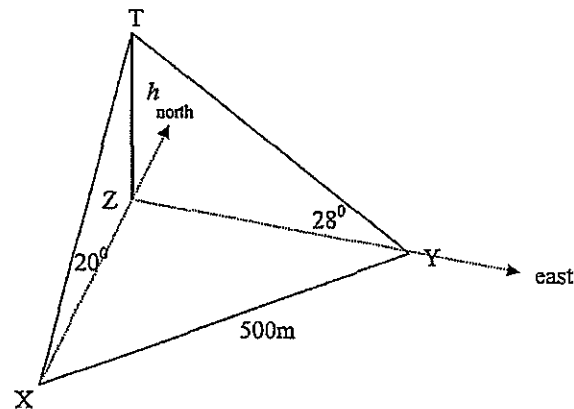
- (ii) Prove that $AC = AT$

2

Question 15 continues on next page

Question 15 continued

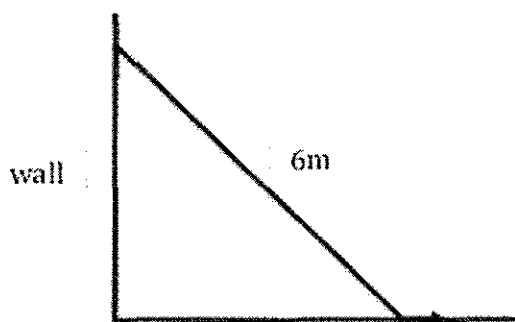
(c)



2

A person observes the angle of elevation of the top of a tree, which is h metres tall, from two positions. From a point X , due south of the tree, it is 20° and from the point Y , due east of the tree, it is 28° . The distance XY is 500m.

(d)



3

A ladder 6 metres long has its upper end against a vertical wall and its lower end on the horizontal floor.

The ladder is initially parallel to the wall, with the lower end at the origin.

The lower end moves away from the wall at a constant speed of 2m/s.

Find the speed at which the upper end moves down the wall two seconds after the lower end has left the wall.

End of paper

MATHEMATICS EXTENSION I – QUESTION

2017 Mathematics Extension I

Multiple choice.

SUGGESTED SOLUTIONS

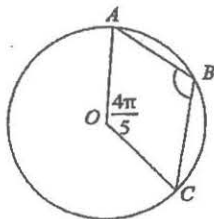
MARKS

MARKER'S COMMENTS

The points A , B and C lie on a circle with centre O , as shown in the diagram.

The size of $\angle AOC$ is $\frac{4\pi}{5}$ radians.

1.



Not to scale

What is the size of $\angle ABC$ in radians?

$$\begin{aligned} \text{reflex angle } \angle AOC &= 2\pi - \frac{4\pi}{5} \\ &= \frac{10\pi}{5} - \frac{4\pi}{5} \\ &= \frac{6\pi}{5} \end{aligned}$$

$$\begin{aligned} \angle ABC &= \frac{1}{2} \times \frac{6\pi}{5} \\ &= \frac{3\pi}{5} \end{aligned}$$

Answer C.

$$2. \int_{\frac{3}{\sqrt{2}}}^3 \frac{4 dx}{\sqrt{9-x^2}}$$

$$= \int_{\frac{3}{\sqrt{2}}}^3 \frac{4 dx}{\sqrt{3^2-x^2}}$$

$$= 4 \left[\sin^{-1}\left(\frac{x}{3}\right) \right]_{\frac{3}{\sqrt{2}}}^3$$

$$= 4 \left[\sin^{-1}\left(\frac{3}{3}\right) - \sin^{-1}\left(\frac{3}{\sqrt{2} \div 3}\right) \right]$$

$$= 4 \left[\sin^{-1}1 - \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \right]$$

$$= 4 \left[\frac{\pi}{2} - \frac{\pi}{4} \right]$$

$$= \pi$$

Answer D.

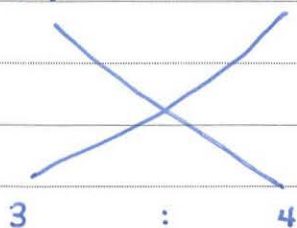
MATHEMATICS EXTENSION I – QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

3. $P(2, 1)$ $Q(2, 8)$



$$= \left[\frac{4(2) + 3(2)}{3+4}, \frac{4(1) + 3(8)}{3+4} \right]$$

$$= \left(\frac{8+6}{7}, \frac{4+24}{7} \right)$$

$$= (2, 4)$$

Answer B.

4.

$$A = \pi r^2$$

$$\frac{dA}{dr} = 2\pi r$$

$$\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$$

$$10 = 2\pi r \cdot \frac{dr}{dt}$$

$$\frac{10}{2\pi r} = \frac{dr}{dt}$$

$$\therefore \frac{dr}{dt} = \frac{5}{\pi r}$$

Answer A.

5. $f(x) = \frac{2}{x-3} + 1$ has $x=3$ & $y=1$ as asymptotes

$f(x)$ has

$\therefore$ the inverse has asymptotes $y=3$ & $x=1$.

Answer A.

MATHEMATICS EXTENSION I – QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$\begin{aligned}
 6. \quad a &= \frac{d}{dx} \frac{1}{2} v^2 \\
 &= \frac{d}{dx} \left[\frac{2x^2 - 3}{2} \right]^2 \\
 &= \frac{d}{dx} \left[\frac{4x^4 - 12x^2 + 9}{2} \right] \\
 &= \frac{d}{dx} [2x^4 - 6x^2 + 4.5] \\
 &= 8x^3 - 12x
 \end{aligned}$$

When $x = 2$

$$\begin{aligned}
 a &= 8(2)^3 - 12(2) \\
 &= 64 - 24 \\
 &= 40 \text{ m/s}^2
 \end{aligned}$$

Answer D.

$$\begin{aligned}
 7. \quad f(x) &= \sin x - \frac{2x}{3} \\
 f'(x) &= \cos x - \frac{2}{3}
 \end{aligned}$$

Formula sheet says

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\begin{aligned}
 f(1.5) &= \sin(1.5) - \frac{2 \times 1.5}{3} \\
 &= \sin(1.5) - 1
 \end{aligned}$$

$$f'(1.5) = \cos(1.5) - \frac{2}{3}$$

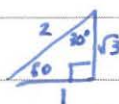
$$x_1 = 1.5$$

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\begin{aligned}
 &= 1.5 - \frac{[\sin(1.5) - 1]}{\cos 1.5 - \frac{2}{3}} \\
 &= 1.496 \quad (3 \text{ dp}).
 \end{aligned}$$

Answer B.

$$\begin{aligned}
 8) \quad \frac{\sqrt{3}}{\sqrt{3}} \tan x &= -\frac{1}{\sqrt{3}} \\
 \tan x &= -\frac{1}{\sqrt{3}}
 \end{aligned}$$



$$\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

formula sheet says...

$$\tan \theta = 2$$

$$\therefore x = n\pi + \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$$

$$\theta = n\pi + \tan^{-1} 2.$$

$$= n\pi - \frac{\pi}{6}.$$

Answer D.

MATHEMATICS EXTENSION I – QUESTION

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

9. $y = 3 \sin^{-1}(2x)$

Background

If $y = \sin^{-1}x$

$$-1 \leq x \leq 1$$

$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

If $y = 3 \sin^{-1}(2x)$

$$-1 \leq 2x \leq 1$$

$$\frac{y}{3} = \sin^{-1}(2x)$$

$$-\frac{1}{2} \leq x \leq \frac{1}{2}$$

$$-\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2}$$

$$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$$

Answer C.

10. $2x^3 - 6x^2 - 8x + 12 = 0$. $a=2$ $b=-6$ $c=-8$ $d=12$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha + \beta + \gamma = \frac{6}{2} = 3$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{-8}{2} = -4$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$\alpha\beta\gamma = \frac{-12}{2} = -6$$

$$(\alpha+2)(\beta+2)(\gamma+2)$$

$$= (\alpha+2)(\beta\gamma + 2\beta + 2\gamma + 4)$$

$$= \alpha\beta\gamma + 2\alpha\beta + 2\alpha\gamma + 4\alpha + 2\beta\gamma + 4\beta + 4\gamma + 8$$

$$= \alpha\beta\gamma + 2(\alpha\beta + \alpha\gamma + \beta\gamma) + 4(\alpha + \beta + \gamma) + 8$$

$$= -6 + 2(-4) + 4(3) + 8$$

$$= -6 - 8 + 12 + 8$$

$$= -14 + 20$$

$$= 6$$

Answer C.

TRIAL EXAM- MATHEMATICS EXTENSION 1 - QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(a) $x - y - 4 = 0$	②	provides correct solution
$3x - y + 4 = 0$		
$y = x - 4$		
$y = 3x + 4$		
$\frac{dy}{dx} = 1$	①	finds gradient of the lines OR
$\frac{dy}{dx} = 3$		
$\therefore m_1 = 1$		shows some understanding.
$\therefore m_2 = 3$		
$\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $		← (NOTE correct formula)
		• wrong formula, no marks!
$= \left \frac{1 - 3}{1 + (1)(3)} \right $		
$= \left \frac{-2}{4} \right $		
$\tan \theta = \frac{1}{2}$		
$\theta = 26.56505118$		
$\therefore \theta = 27^\circ$		
(b) $y = \ln(\sin^{-1} x)$	②	provides correct solution
$\frac{dy}{dx} = \frac{1}{\sin^{-1} x} \times \frac{1}{\sqrt{1-x^2}}$		
$= \frac{1}{\sin^{-1} x \sqrt{1-x^2}}$	①	demonstrates understanding of differentiating a log, ie $\frac{1}{f(x)} \times f'(x)$
• Correct rule on Reference sheet!	where $f(x) = \sin^{-1} x$ and $f'(x) = \frac{1}{\sqrt{1-x^2}}$	

TRIAL EXAM- MATHEMATICS EXTENSION 1 – QUESTION 1

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(e) let $P(x) = x^3 - 3x^2 - 10x + 24$		
$P(-3) = (-3)^3 - 3(-3)^2 - 10(-3) + 24$		
$= -27 - 27 + 30 + 24$		
$= 0$		
Since $P(-3) = 0$ then $(x+3)$ is a factor of $P(x)$.	①	"show that" with working.
$x^2 - 6x + 8$ — ① or equivalent		
$ \begin{array}{r} x+3 \overline{) x^3 - 3x^2 - 10x + 24} \\ \underline{-(x^3 + 3x^2)} \\ -6x^2 - 10x \\ \underline{-(-6x^2 - 18x)} \\ 8x + 24 \\ \underline{-(8x + 24)} \\ 0 \end{array} $		
$\therefore P(x) = (x+3)(x^2 - 6x + 8)$		
$= (x+3)(x-4)(x-2)$	① ①	for each factor
• If the question says "show that", then it is asking you to <u>completely justify</u> the given answer <u>by showing every step of working</u> .		

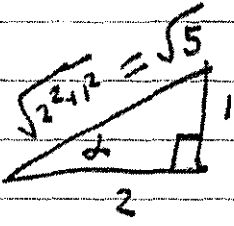
TRIAL EXAM- MATHEMATICS EXTENSION 1 – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
<u>OR</u> (e)		
Let $P(x) = x^3 - 3x^2 - 10x + 24$		
$x^2 - 6x + 8$ — ① or equivalent		
$ \begin{array}{r} x+3 \overline{) x^3 - 3x^2 - 10x + 24} \\ \underline{- x^3 + 3x^2} \downarrow \\ -6x^2 - 10x \downarrow \\ \underline{-6x^2 - 18x} \downarrow \\ 8x + 24 \\ \underline{- 8x + 24} \\ 0 \end{array} $		
Since there is a zero remainder,		
then $(x+3)$ is a factor of $P(x)$. — ① for "show		
		that" with
		working
$ \begin{aligned} P(x) &= (x+3)(x^2 - 6x + 8) \\ &= (x+3)(x-4)(x-2) \end{aligned} $		
① ① for each factor.		

MATHEMATICS EXTENSION I – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) $\int_{-1}^0 x\sqrt{1+x} dx = \int_0^1 (u-1)\sqrt{u} du$ $u = 1+x$ $\therefore x = u-1$ $\frac{dx}{du} = 1$ $dx = du$ when $x = -1, u = 0$ when $x = 0, u = 1$ $= \int_0^1 u\sqrt{u} - \sqrt{u} du$ $= \int_0^1 u^{\frac{3}{2}} - u^{\frac{1}{2}} du$ $= \left[\frac{2u^{\frac{5}{2}}}{5} - \frac{2u^{\frac{3}{2}}}{3} \right]_0^1$ $= \frac{2}{5} - \frac{2}{3}$ $= -\frac{4}{15}$	1 for correct substitution 1 for correct limits of integration 1	1 for correct substitution 1 for correct limits of integration
b) $\frac{2x}{x-1} \geq 1 \quad x \neq 1$ $(x-1)^2 \frac{2x}{x-1} \geq (x-1)^2$ $2x(x-1) \geq (x-1)^2$ $2x(x-1) - (x-1)^2 \geq 0$ $(x-1)[2x - (x-1)] \geq 0$ $(x-1)(x+1) \geq 0$	1 1	
Sketch $y = (x-1)(x+1)$ $\therefore -1 \leq x, x > 1$	1	Recognising that $x \neq 1$ was an integral part of this question; you could not earn the 3rd mark without it

MATHEMATICS EXTENSION I – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) $\sin(2 \tan^{-1} \frac{1}{2})$ let $\alpha = \tan^{-1} \frac{1}{2}$ $\therefore \tan \alpha = \frac{1}{2}$  $\sin(2 \tan^{-1} \frac{1}{2}) = \sin 2\alpha$ $= 2 \sin \alpha \cos \alpha$ $= 2 \times \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}}$ $= \frac{4}{5}$	1 1	
d) i $\ddot{x} = 0$ $\dot{x} = \int 0 dt$ $= C_1$ when $t=0$, $\dot{x} = 50$ $50 = C_1$ $\therefore \dot{x} = 50$ $x = \int 50 dt$ $= 50t + C_2$ when $t=0$, $x=0$ $0 = 50(0) + C_2$ $C_2 = 0$ $\therefore x = 50t$	1	Note that your working <u>must</u> begin at $\ddot{x}=0$, and the constant of integration <u>must</u> be calculated at each step.
$\ddot{y} = -g$ $\dot{y} = \int -g dt$ $= -gt + C_3$ when $t=0$, $\dot{y}=0$ $0 = -g(0) + C_3$		

MATHEMATICS EXTENSION I – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$\therefore C_3 = 0$ $\therefore \dot{y} = -gt$ $y = \int -gt \, dt$ $= -\frac{1}{2}gt^2 + C_4$ when $t=0, y=90$ $\therefore 90 = -\frac{1}{2}g(0)^2 + C_4$ $\therefore C_4 = 90$ $\therefore y = -\frac{1}{2}gt^2 + 90$ $= 90 - \frac{1}{2}gb^2$	1	
ii The ball strikes the ground when $y=0$ $90 - \frac{1}{2}gt^2 = 0$ $\frac{1}{2}gt^2 = 90$ $gt^2 = 180$ $t^2 = \frac{180}{g}$ $t = \pm \sqrt{\frac{180}{g}}$ $= \sqrt{36} \times \sqrt{\frac{5}{g}} \quad (t > 0, \text{ time})$ $t = 6\sqrt{\frac{5}{g}}$	1	
iii When $t = 6\sqrt{\frac{5}{g}}$, $x = 50 \times 6\sqrt{\frac{5}{g}}$ $= 300\sqrt{\frac{5}{g}}$ $\therefore$ lands $300\sqrt{\frac{5}{g}}$ metres from tower	1	

MATHEMATICS EXTENSION I - QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(a) i) $\sqrt{3} \sin t + \cos t \equiv R \sin(t + \alpha)$ $= R \sin t \cos \alpha + R \cos t \sin \alpha$ $= R \cos \alpha \cdot \sin t + R \sin \alpha \cdot \cos t$ Equating coefficients $R \cos \alpha = \sqrt{3} \quad \text{--- (1)}$ $R \sin \alpha = 1 \quad \text{--- (2)}$ $(1)^2 + (2)^2$ $R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = (\sqrt{3})^2 + 1$ $R^2 (\cos^2 \alpha + \sin^2 \alpha) = 3 + 1$ $R^2 = 4$ $R = 2 \quad \text{since } R > 0$ and $(2) \div (1)$ $\frac{\sin \alpha}{\cos \alpha} = \frac{1}{\sqrt{3}}$ $\tan \alpha = \frac{1}{\sqrt{3}}$ $\alpha = \frac{\pi}{6}$ $\therefore \sqrt{3} \sin t + \cos t = 2 \sin(t + \frac{\pi}{6})$	1	This part of the question was mostly done well.
ii) From (1) $2 \sin(t + \frac{\pi}{6}) = \sqrt{3}$ $\sin(t + \frac{\pi}{6}) = \frac{\sqrt{3}}{2}$ $t + \frac{\pi}{6} = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{13\pi}{3}, \dots$ $t = \frac{\pi}{6}, \frac{\pi}{2}, \frac{25\pi}{6}, \dots$ but $0 \leq t \leq 2\pi$ $\therefore t = \frac{\pi}{6} \text{ or } \frac{\pi}{2}$	1	Some students did not consider the domain that $0 \leq t \leq 2\pi$.

MATHEMATICS EXTENSION I – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$b) \quad V = \pi \int_0^{\pi} 9 \cos^2 \frac{x}{2} dx$	1/2	
$= \pi \int_0^{\pi} 9 \cos^2 \frac{x}{2} dx$		
$= 9\pi \int_0^{\pi} \frac{1}{2} (1 + \cos x) dx$	1	
$= \frac{9\pi}{2} \int_0^{\pi} 1 + \cos x dx$		
$= \frac{9\pi}{2} \left[x + \sin x \right]_0^{\pi}$	1/2	
$= \frac{9\pi}{2} \left[(\pi + \sin \pi) - (0 + \sin 0) \right]$	1/2	
$= \frac{9\pi^2}{2}$	1/2	(3)
$c) i) \quad \text{Let } T = T_0 + Ae^{-kt} \quad \text{--- (1)}$		
$\frac{dT}{dt} = -kAe^{-kt}$	1	
$= -k[Ae^{-kt} + T_0 - T_0]$	1	(2)
$= -k[T - T_0] \quad \text{from (1)}$		
$\text{OR} \quad \text{From (1)} \quad T - T_0 = Ae^{-kt}$	1	
$\frac{dT}{dt} = -kAe^{-kt}$	1	
$= -k[T - T_0] \quad \text{from (2)}$		
T_0		Students who did not show where Ae^{-kt} came from received only 1 mark.

MATHEMATICS EXTENSION I – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d.ii) At $t=0$, $T_0=12$, $T=60$ $t=25$, $T=30$ $t=?$, $T=15^\circ\text{C}$ Using $T = T_0 + Ae^{-kt}$ when $t=0$, $T_0=12$, $T=60$ $60 = 12 + Ae^{-k(0)}$ $48 = Ae^0$ $A = 48$ $\therefore T = 12 + 48e^{-kt}$ when $t=25$, $T=30$ $30 = 12 + 48e^{-k(25)}$ $18 = 48e^{-25k}$ $\frac{3}{8} = e^{-25k}$ $-25k = \ln\left \frac{3}{8}\right $ $k = -\frac{\ln\left \frac{3}{8}\right }{25} \quad \text{--- (1)}$ when $T=15$, $=kt$ $15 = 12 + 48e^{-kt}$ $3 = 48e^{-kt}$ $\frac{1}{16} = e^{-kt}$ $-kt = \ln\left \frac{1}{16}\right $ $t = \frac{\ln\left \frac{1}{16}\right }{-k}$ $= \frac{\ln\frac{1}{16}}{\frac{\ln\left \frac{3}{8}\right }{25}} = 70.66 \dots$ $\frac{\ln\left \frac{3}{8}\right }{25} = \frac{0}{25} \text{ 71 minutes!}$	1 1 1	(3) Some student rounded down to 70 minutes when it should have been rounded up to 71 min. Marks were still awarded

MATHEMATICS EXTENSION I - QUESTION 14

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

(a) $x^2 = 4ay$

1.

some students

(1) $y = \frac{x^2}{4a}$

found $\frac{dx}{dt}$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

and $\frac{dy}{dt}$

$$= \frac{x}{2a}$$

used $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ When $x = 2at$

$$\frac{dy}{dx} = \frac{2at}{2a}$$

Well done.

$$= t. \quad \leftarrow \text{must prove}$$

 $\therefore$ gradient of tangent at $(2at, at^2)$ is t .equation of tangent at $(2at, at^2)$

$$y - y_1 = m(x - x_1)$$

$$y - at^2 = t(x - 2at)$$

$$y - at^2 = tx - 2at^2$$

$$y = tx - 2at^2 + at^2$$

$$y = tx - at^2$$

(ii) line d parallel to tangent at P through focus
for parallel lines $m_1 = m_2$

1

. Some students
incorrectly
stated the focus.

$$\therefore m = t$$

Focus $(0, a)$

$$y - y_1 = m(x - x_1)$$

$$y - a = t(x - 0)$$

. well done.

$$y - a = tx$$

$$y = tx + a$$

MATHEMATICS EXTENSION I – QUESTION 14

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
pg 2 Q 14		
iii) $y = tx + a$ for x intercept, $y = 0$. $0 = tx + a$ $tx = -a$ $x = -\frac{a}{t}$		well done.
$\therefore R(-\frac{a}{t}, 0) \quad S = (0, a)$	R(1)	
$M = \text{Midpoint}_{RS}$ $= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$ $= \left(\frac{-\frac{a}{t} + 0}{2}, \frac{0 + a}{2} \right)$ $= \left(-\frac{a}{2t}, \frac{a}{2} \right)$	M(1)	0
iv) $M = \left(-\frac{a}{2t}, \frac{a}{2} \right)$ $x = -\frac{a}{2t}$ $y = \frac{a}{2}$ $\therefore y$ is independent of x. x has a domain of all x except $x=0$ $\therefore$ equation of locus is $y = \frac{a}{2} (x \neq 0)$. $x=0$, (this is because if at vertex, line through focus parallel to tangent will have no x intercept.)	(1)	only 1 student noted $x \neq 0$.

MATHEMATICS EXTENSION I – QUESTION

14	SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(b)	$\cos(2 \cos^{-1} x)$		
	let $\alpha = \cos^{-1} x$	1	difficulties encountered
	$\cos \alpha = x$		using
			$\cos 2\alpha = 1 - 2\sin^2 \alpha$
	$\therefore \cos(2 \cos^{-1} x) = \cos 2\alpha$		etc
	$= 2 \cos^2 \alpha - 1$		
	$= 2x^2 - 1$		
	hence...		
	$\int_0^{\frac{1}{2}} \cos(2 \cos^{-1} x) dx$	1	
	$= \int_0^{\frac{1}{2}} (2x^2 - 1) dx$		
	$= \left[\frac{2x^3}{3} - x \right]_0^{\frac{1}{2}}$		
	$= \frac{1}{12} - \frac{1}{2} = 0$		
	$= -\frac{5}{12}$		
			Students needed to
(i)	$x = 1 + 2 \cos(2t - \frac{\pi}{3})$	1.	Show <u>all</u>
	$\dot{x} = -4 \sin(2t - \frac{\pi}{3})$		steps for
	$\ddot{x} = -8 \cos(2t - \frac{\pi}{3})$		full marks
	$= -8 [2 \cos(2t - \frac{\pi}{3}) + 1 - 1]$		
	$= -4 [\underbrace{2 \cos(2t - \frac{\pi}{3}) + 1}_{x-1}]$		
	since $x = 1 + 2 \cos(2t - \frac{\pi}{3})$		
(ii)	Centre of motion. $\ddot{x} = 0$	1.	
	$0 = -4(x-1)$		

$\therefore x = 1$ is the centre of the motion

$$\therefore \sin\left(2t - \frac{\pi}{3}\right) = \pm 1$$

$$2t - \frac{\pi}{3} = \frac{\pi}{2} \pm n\pi \quad n \text{ integer}$$

$$2t = \frac{5\pi}{6} \pm n\pi$$

$$t = \frac{5\pi}{12} \pm \frac{n\pi}{2}$$

$\therefore$ First time is when $n=0$

$\therefore$ First time max^m speed is reached is $t = \frac{5\pi}{12}$ seconds

(iii) if at rest, $v=0$. $t=?$ First time $v=0$.

$$x = -4 \sin\left(2t - \frac{\pi}{3}\right)$$

$$0 = \sin\left(2t - \frac{\pi}{3}\right)$$

$$\therefore 2t - \frac{\pi}{3} = n\pi \quad n \text{ integer}$$

$$2t = \frac{\pi}{3} + n\pi \quad n \text{ integer}$$

$$t = \frac{\pi}{6} + \frac{n\pi}{2}$$

$\therefore$ First time will be when $n=0$

$$\therefore t = \frac{\pi}{6} \text{ seconds} \quad (1)$$

$$\text{amplitude} = \underline{2} \quad (1)$$

MATHEMATICS EXTENSION I - QUESTION 15

①

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) Show that $5^n > 4^n + 3^n$ by mathematical induction.		
<u>Step 1</u>		
Show that the result is true for $n=3$	1/2	The majority of students got this mark.
$5^3 > 4^3 + 3^3$		
LHS = 5^3		
= 125		
RHS $4^3 + 3^3$		
= 91		
Since $125 > 91$		
$5^n > 4^n + 3^n$ for $n=3$		
<u>Step 2</u>		
Assume the result is true for $n=k$	1/2	Well done
(an integer ≥ 3)		
i.e. Assume $5^k > 4^k + 3^k$		
<u>Step 3</u>		
Prove the result is true for $n=k+1$, assuming the statement is true for $n=k$	1/2	The majority of students know the correct layout for mathematical induction problems.

MATHEMATICS EXTENSION I – QUESTION

②

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
Prove $5^{k+1} > 4^{k+1} + 3^{k+1} \rightarrow *$ if $5^k > 4^k + 3^k$ LHS = 5^{k+1} = $5(5^k)$ > $5(4^k + 3^k)$ using the assumption = $5 \cdot 4^k + 5 \cdot 3^k$ > $4 \cdot 4^k + 3 \cdot 3^k$ = $4^{k+1} + 3^{k+1}$ $\therefore 5^{k+1} > 4^{k+1} + 3^{k+1}$	$\frac{1}{2}$ $\frac{1}{2}$	This part of the proof was not attempted as expected. Common mistakes made by students include: • Modifying the statement $*$ without using the assumption. • Using the inequality as an equation by substituting $5^k = 4^k + 3^k$ directly into the statement $*$.
<u>Step 4</u> Since the result is true for $n=3$, it is also true for $n=k+1$, i.e., $n=3+1$, and so on for $n=3, 4, 5, \dots$ Therefore, by the principle of Mathematical Induction it holds true for all $n \geq 3$	$\frac{1}{2}$	

MATHEMATICS EXTENSION I - QUESTION

③

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
i) $\angle ABT = \pi/2 - \theta$		
In $\triangle ABC$, $AB = AC$ (tangents from an external common point to a circle are equal)	$\frac{1}{2}$	students were not penalised for not using the
$\therefore \angle ABC = \angle ACB = \theta$ (angles opposite equal sides of a triangle are equal)	$\frac{1}{2}$	appropriate terminology for the circle
In $\triangle BCD$, $\angle CBD = \pi/2$ (angle subtended by the diameter is a right angle).	$\frac{1}{2}$	geometry theorems but it is highly
$\therefore \angle CBT = \pi/2$ ($\angle DBT$ is a straight angle)	$\frac{1}{2}$	recommended that
$\therefore \angle ABT = \pi/2 - \theta$		everyone re-visits these theorems.
		For example, the majority of students used
		an alternate segment theorem in short.
		students using alternative methods

MATHEMATICS EXTENSION I – QUESTION

(4)

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
		were awarded
		1/2 mk for
		each
		theorem.
		leading
		towards
		the proof.
c) $\tan 20^\circ = \frac{h}{xz}$		Quite
$xz = \frac{h}{\tan 20^\circ}$ or $\cot 20^\circ$		Well done
$\tan 28^\circ = \frac{h}{yz}$	1/2	by the
$yz = \frac{h}{\tan 28^\circ}$ or $\cot 28^\circ$		majority
$xz^2 + yz^2 = 500^2$ Pythagoras' Theorem.		of students
$\left(\frac{h}{\tan 20^\circ}\right)^2 + \left(\frac{h}{\tan 28^\circ}\right)^2 = 500^2$	1/2	
$h^2 \left[\left(\frac{1}{\tan 20^\circ}\right)^2 + \left(\frac{1}{\tan 28^\circ}\right)^2 \right] = 500^2$		
$h^2 \left[\frac{\tan^2 25 + \tan^2 20}{\tan^2 20^\circ \tan^2 28^\circ} \right] = 250000$		Some
$h^2 = \frac{250000 \times \tan^2 20^\circ \times \tan^2 28^\circ}{\tan^2 20^\circ + \tan^2 28^\circ}$	1/2	students
$= \sqrt{22551.44}$		made
$= 150 \text{ metres (nearest metre)}$	1/2	mistakes
		in their
		calculations
		at this
		stage.

MATHEMATICS EXTENSION I – QUESTION

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) From Pythagoras, $x^2 + y^2 = 36$ $y = (36 - x^2)^{1/2}$		
$\frac{dy}{dx} = \frac{1}{2} (36 - x^2)^{-1/2} \times -2x$	$\frac{1}{2}$	
$= \frac{-x}{\sqrt{36 - x^2}}$	$\frac{1}{2}$	
From the chain rule, and given that the rate at which the bottom of the ladder slides to the right $\frac{dx}{dt}$ is 2 m/s		
$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$		
$= \frac{-x}{\sqrt{36 - x^2}} \times 2$		
$= \frac{-2x}{\sqrt{36 - x^2}}$	$\frac{1}{2}$	
When $t = 2$, $x = 4$ m	$\frac{1}{2}$	
$\frac{dy}{dt} = \frac{-8}{\sqrt{36 - 16}}$	$\frac{1}{2}$	
$= \frac{-8}{\sqrt{20}}$		Some students left the speed as
$= \frac{-4}{\sqrt{5}} \text{ m/s}$	$\frac{1}{2}$	$\frac{-4}{\sqrt{5}}$ No
$\therefore$ the speed at which the top falls is $\frac{4}{\sqrt{5}} \text{ m/s}$		marks were deducted: speed, being a scalar quantity should be left as positive.